Harmonization of Multispectral and Panchromatic Images From VHR/EHR Spaceborne Scanners

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***Abstract*—This study presents a new concept related to the quality and usability of multispectral (MS) and panchromatic (PAN) datasets captured by very/extremely high-resolution (VHR/EHR) satellite MS scanners. Despite the ever-increasing resolution and spectral coverage capabilities of modern instruments, MS datasets suffer from the presence of aliasing originating from the insufficient sampling step size, corresponding to a high value of the modulation transfer function (MTF) of the instrument at the Nyquist frequency. Another source of degradation is the local misalignment between interpolated MS and PAN, originating from different viewpoints along the orbit of the instruments in the presence of discontinuities of the imaged surface. Here, we propose a procedure to harmonize MS and PAN datasets: aliasing artifacts of MS are suppressed; local MS-to-MS and MS-to-PAN shifts are damped. This is feasible for the unimodality of the MS and PAN instruments. Harmonization of MS toward PAN is achieved by injecting the least-squares (LS) residue of the multivariate linear regression of the interpolated bands captured by each MS scanner toward the low-pass-filtered PAN. Experiments with two (MS-PAN) and three (MS1-MS2-PAN) onboard instruments show that an almost perfect alignment and a significant reduction of aliasing patterns are achieved. MS pansharpening greatly benefits from harmonized datasets. When the latter are used for the fusion and as indicators of spectral and spatial qualities of fusion products, inconsistencies noticed and complained about in the literature vanish. The procedure is in real time, fully automated, and does not require additional or ancillary information.**

***Index Terms*—Aliasing, coregistration, GeoEye-1/2, multispectral (MS) pansharpening, multivariate regression, parallaxes, satellite remote sensing, WorldView-2/3.**

I. Introduction

SINCE the pioneering very high-resolution (VHR) imager IKONOS-2, launched in 1999, modern multispectral (MS) scanners mounted on satellite platforms have reached submetric resolutions in the panchromatic (PAN) wavelengths. The present generation of WorldView satellites for Earth observation (EO) features extremely high-resolution (EHR) acquisition capability: up to 1.24 m for MS and 0.31 m for PAN, at nadir. The capability of resolving less than 1 m from a height of about 650 km is achieved at the cost of extremely cumbersome and heavy optics. Constraints in the design of the system have motivated the use of a unique telescope for the MS and PAN instruments [1]. The MS bands having lower spatial resolution are acquired with a relatively high value of the modulation transfer function (MTF) at the Nyquist frequency, that is, half the sampling frequency. The PAN image is sampled at a frequency typically four times larger and postprocessed at the ground station for MTF restoration. The presence of parallaxes during the acquisition of the individual instruments and the presence of aliasing impairments in the MS bands, but not in the PAN image, constitute a sort of discrepancy in the bundle datasets. The nonconformity of the data is the main reason for possible decrements in performance whenever application tasks, e.g., fusion, classification, and detection, are pursued.

In this study, we present two totally unsupervised procedures for spatial *harmonization*, i.e., aliasing reduction and alignment of the interpolated MS bands to the PAN image. The former is suitable for systems featuring a unique instrument for all MS bands, e.g., GeoEye-1 and Pléiades, as well as GeoEye-2, which ceased operations in 2019, IKONOS-2 and QuickBird-2, both dismissed in 2015. The latter is suitable for the two MS instruments of WorldView-2/3. This is possible because the PAN channel is statistically correlated with the MS bands. The fact that a unimodal fusion, as pansharpening, was potentially capable of recovering aliasing and misalignment between the MS and PAN datasets was recognized by two of the authors several years ago [2]. However, until 2020, no alignment in spectral diversity (color hues) was feasible, only of spatial structures [3]. A simultaneous approach is totally different because it is based on the model of a specific instrument [4] and works with four bands + PAN. Another valuable effort works with submetric PAN [5] but requires considerable processing resources because it explicitly deploys numerical techniques of convex optimization to emulate what is implicitly done by some pansharpening methods. The proposed harmonization method is real-time, fully automated, and may be used for any satellite system and, in principle, extended to more than three onboard instruments, e.g., WorldView-3 featuring a fourth onboard instrument.

This article is organized in the following way. Section II investigates the causes of the mismatch between the MS and PAN datasets. Section III lays the foundations of the harmonization procedure and defines two versions, one suit-

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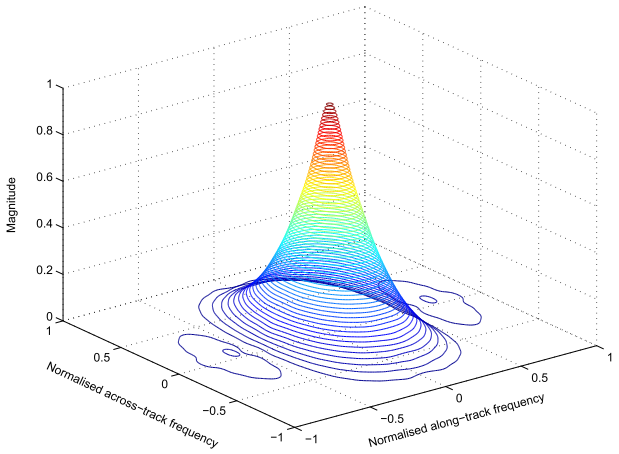


Fig. 1. Laboratory MTF of a spectral channel of the Pléiades instrument.

able for most systems, featuring two separate instruments, one for MS and one for PAN, and another suitable for WorldView-2/3, having two separate visible and near infrared (VNIR) scanners. Section IV reviews the rationale for the fusion assessment, focusing on full-scale evaluations. Section V presents extensive results on residue analysis, correction of aliasing and shifts, and pansharpening for the GeoEye-1 and WorldView-2 datasets. Section VI discusses the main issues encountered and explains the advantages of the harmonization procedure. Section VII suggests possible application scenarios and developments.

II. Materials

This section provides an in-depth explanation of the causes that generate discrepancies among the different image datasets, more than two, in principle. Fig. 1 illustrates the laboratory MTF of one of the spectral channels of the MS instrument Pléiades. Note that the spatial frequency response in the along- and across-track directions is squeezed along the track due to the motion of the platform [6]. Due to diffraction of the optics, the MTF is shrunk as the wavelength increases. Thus, the spatial resolution is slightly better for the leftmost part of the spectrum, corresponding to the violet color [7].

The presence of a high amount of signal in the passband of the instrument ensures an adequate visual sharpness but entails the onset of aliasing artifacts due to insufficient sampling step size. Conversely, the PAN image, having four times lower step size and thus four times greater Nyquist frequency, for an MS-to-PAN scale ratio equal to four, is acquired with a much lower value at Nyquist of MTF, which is roughly the same for all spectral channels. However, a high signal-to-noise ratio (SNR) is achieved, owing to the time-delayed integration (TDI), which is directly implemented on the charge-coupled device (CCD). The PAN image that is captured, being very little noisy and practically aliasing-free, is then postprocessed at the ground station for restoration of the MTF. An edge-crispening filter, that is, an all-pass filter plus a high-pass filter, is applied during the geocoding process, which includes resampling and interpolation. The characteristics of the MTF, having the shape of a Chinese hat with heavy flaps, allow typical values at Nyquist frequency around 0.3 for the MS channels and less than 0.1, on orbit, that is, before restoration, for the PAN channel, to be simultaneously achieved. After restoration, the MTF of the geocoded PAN is approximately equal to that of the original MS. Upon these premises, the PAN image, both on-orbit and geocoded, is practically aliasing-free because the edge-crispening filter does not affect aliasing. The MS bands are always marred by annoying jagged artifacts due to aliasing, especially noticeable in the presence of elongated geometric structures. The question that arises is the following: is it possible to get aliasing-free MS bands? The answer is yes [8] because: 1) the two sensors are unimodal (the MS and PAN instruments share the same optics and differ only in spectral and spatial sampling) and 2) the PAN image is aliasing-free.

In parallel to the spatial resolution, the number of spectral bands of new-generation MS scanners has been increased to finely cover the VNIR wavelengths. Fig. 2 shows the spectral responsivity channels of a typical four-band MS + PAN system (GeoEye-1) and the nine spectral channels (4 + 4 + PAN) of WorldView-2.

Upon these premises, the MS and PAN instruments sharing the same optics and thus the same focal plane cannot have strictly parallel axes, but there is a small angle contained in the orbit plane. The immediate consequence is that the two/three instruments acquire different areas at the same instant or, equivalently, they scan the same area at slightly different times. Thus, moving objects in the scene, e.g., vehicles, appear displaced in the MS and in the PAN images. The GeoEye-1 MS instrument simultaneously takes all the MS bands in a bundle under the same perspective. Therefore, the MS bands may be rigidly shifted from one another, a type of misalignment that is easily corrected before the data are delivered but are not locally warped by an extent variable on the height of terrain.

The situation is totally different for the PAN instrument, which images the same scene as the MS instrument does, at a different time, and hence from a different viewpoint along the orbit. In the presence of terrain relief, the same object is imaged at different positions in the MS and PAN images. Fig. 3(a) illustrates the acquisition of two separate instruments. The small angle between the axes of the two instruments is magnified along the orbit for display convenience. The difference in positions of the same point acquired from different viewpoints, or *parallaxes*, is called *disparity* and depends on the height of the point itself. While in stereo vision, the disparity is measured and used to estimate the unknown height, in order to align the same point framed in the two images, its ground height must be previously known. This operation, inverse to stereo restitution, is performed by orthorectification, whose goal is to produce an image taken from the nadir at infinity distance, which can be exactly superimposed onto a cartographic map. After the MS and PAN images have been orthorectified, they can be perfectly overlapped with a simple rigid shift. However, the precision of orthorectification depends on the spatial resolution of the digital elevation model (DEM) used to enforce its calculation [9], [10]. If the spatial sampling and/or the height accuracy of the DEM is too coarse

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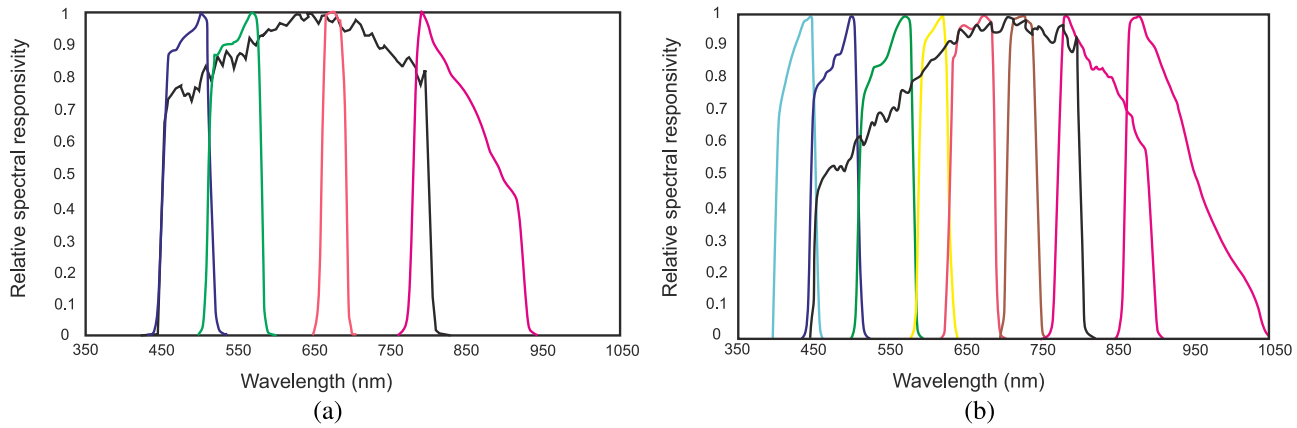


Fig. 2. Spectral responsivity functions of (a) GeoEye-1, bands blue, PAN, green, red, and NIR, from left to right and (b) WorldView-2, bands coastal, blue, PAN, green, yellow, RE, NIR1, and NIR2, with one MS scanner (MS1) taking the same four bands as GeoEye-1; another MS scanner (MS2) capturing coastal, yellow, RE, and NIR2.

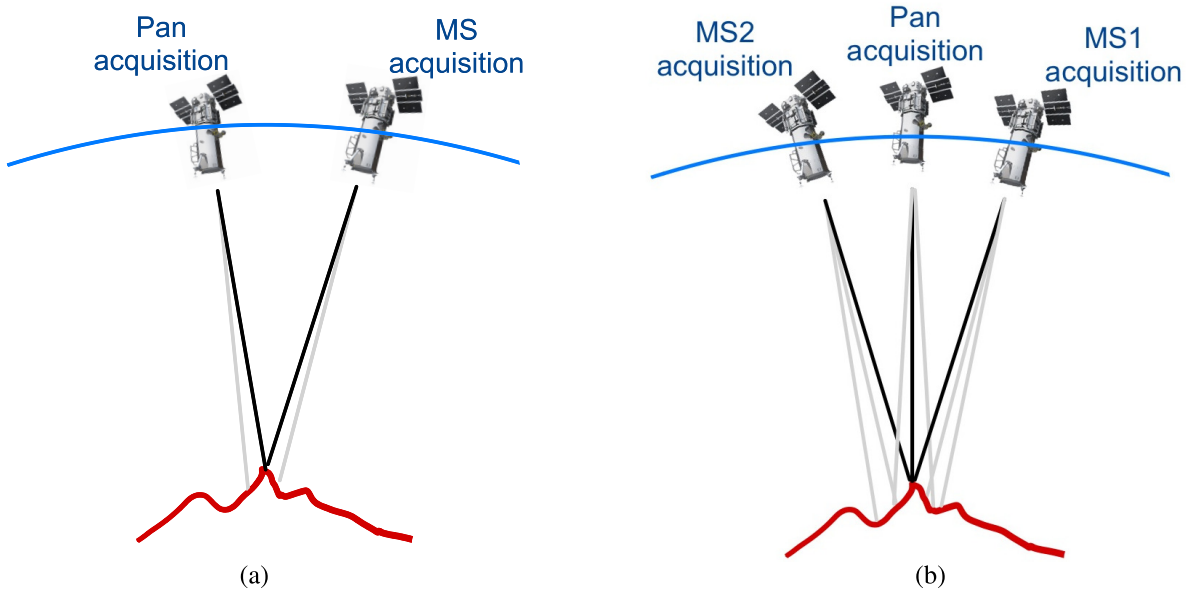


Fig. 3. Acquisition geometries of MS and PAN satellite scanning systems. (a) GeoEye-1 features two separate instruments, one for the four MS bands and another for PAN, whose optical axes are not parallel. The acquisition of the same ground point occurs on different instants, hence from different points along the orbit. (b) WorldView-2/3 features two distinct MS scanners, MS1 and MS2, in addition to PAN. The optical axis of the instrument that performs the acquisition from the respective position along the orbit is shown with a solid dark line. All orbital acquisition angles are deliberately exaggerated.

with respect to the pixel size, the height-driven local warping is insufficient, and a residual disparity survives. Another cause of malfunction is that the DEM usually represents the height of the bare soil and not of possible natural (plants and trees) and man-made (buildings, bridges, and viaducts) structures.

The situation described before is even more complicated for satellites equipped with more than two instruments, such as the WorldView-2/3 platforms, in which eight MS bands are taken by two separate instruments. In WorldView-2, three instruments are present on the platform: the classical four-band MS instrument (MS1) and the new MS instrument (MS2), featuring four more bands, partially interleaved with the others, as shown in Fig. 2(b). WorldView-3 is also equipped with a fourth instrument, namely, a short-wave infrared (SWIR) scanner, although the latter will not be considered here. The three instruments acquire different areas at the same instant or, equivalently, they scan the same area at slightly different times. Consequently, moving objects in the scene appear not only shifted in the two sets of MS bands and the PAN images [11], but the two sets of MS bands and the PAN image are also captured at different moments from distinct positions along the orbit; hence, under different viewing perspectives, with the consequence that the two sets of MS bands may be locally shifted to one another. Fig. 3(b) illustrates the acquisition geometry of the WorldView-2 platform. For the sake of clarity, the angles between the optical axes have been intentionally exaggerated. Now, a question arises: is it possible to correct local misalignment due to parallax view between the MS bands and PAN and between MS bands captured by different instruments? Analogously to what concerns aliasing,

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the answer is yes, because notwithstanding there are two or more than two MS instruments, there is a unique PAN image.

III. Methods

The term harmonization of MS toward PAN indicates the process by which MS is forced to be spatially consistent with a low-pass version of PAN having the same spatial frequency content as MS. PAN is the target, and spatial artifacts of MS that are missing in PAN, like aliasing, should be removed or at least mitigated. Local or global shifts between resampled MS and PAN should be damped, and the former should become perfectly overlapped with the latter. While shifts might in principle be corrected by resorting to more finely spaced DEMs, aliasing artifacts are unavoidable, unless all MS bands are taken at higher resolution with TDI, the same as PAN, which can be eliminated. However, there would be as many separate onboard instruments as the MS bands, and this would create serious problems with parallaxes.

So far, coregistration has usually been assumed between the MS and PAN datasets. Whenever this does not happen, an inaccurate MS-to-PAN overlap can have devastating effects on the quality of products. When pansharpening fusion is concerned, methods based on component substitution (CS) tend to damp small misalignments and prevent the geometry of the scene from fading after fusion [8], [12]. Depending on the injection model, there may be a small residual color change, but not in the structures [2], [3]. Conversely, the class of methods based on multiresolution analysis (MRA) is extremely sensitive to spatial misalignment that causes a loss of synchronization of the low-pass and high-pass spatial frequency components of sharp edges and structures and their consequent fading. The spectral-spatial duality of the classes of CS and MRA pansharpening methods makes the latter robust to spectral impairments, that is, MS and PAN datasets taken from different satellites and/or at different times [13]. This property is invaluable for multiplatform fusion [14]. Generally speaking, the fusion of heterogeneous datasets (multimodal fusion), such as thermal and optical datasets, cannot be approached as a problem of CS [15].

For a unimodal fusion, as pansharpening, a multivariate linear regression between the low-pass-filtered PAN, $\mathbf{P}_L$, and the interpolated MS bands, $\widetilde{\mathbf{M}}_k$, where the phase of the linear interpolating kernel should be matched to the acquisition geometry to avoid introducing systematic shifts [16], has been used to model the spectral and spatial relationships among the different channels of the instrument [17]

$$\mathbf{P}_L = \widehat{w}_0 + \sum_{k=1}^{N} \widehat{w}_k \cdot \widetilde{\mathbf{M}}_k + \boldsymbol{\epsilon} \triangleq \widehat{\mathbf{I}}_L + \boldsymbol{\epsilon} \tag{1}$$

in which $\widehat{\mathbf{I}}_L$ is the matching intensity component and ϵ the minimum-variance, space-varying residue, having mean equal to zero. If we set $\widehat{\boldsymbol{w}} = \{\widehat{w}_k\}_{k=0,\ldots,N}$, the least-squares (LS) solution of (1) corresponds to the minimization of the Frobenius norm, $\|\cdot\|_F^2$, between the low-pass-filtered PAN, $\mathbf{P}_L$, and the multivariate linear regression of the interpolated MS bands

$$\widehat{\boldsymbol{w}} = \arg\min_{\boldsymbol{w}} \left\{ \left\| \mathbf{P}_L - w_0 - \sum_{k=1}^{N} w_k \cdot \widetilde{\mathbf{M}}_k \right\|_F^2 \right\}. \tag{2}$$

A measure of the success of the matching of (1) is given by the coefficient of determination, R^2, defined as follows:

$$R^2 \triangleq 1 - \frac{\sigma_\epsilon^2}{\sigma_{\mathbf{P}_L}^2} \tag{3}$$

in which σ_ϵ^2 and $\sigma_{\mathbf{P}_L}^2$ denote the variance of the LS residue and the low-pass-filtered PAN image, respectively.

A. Harmonization of GeoEye-1 Data (Four Bands + PAN)

In the seminal article on the spatial properties of the MS pansharpening methods [8], the shift tolerance of the CS-based fusion methods has been mathematically proven. However, no hint was presented of how to exploit these properties in contexts other than those of fusion. It was also unclear how to extend the shift invariance to the spectral component (color hues) and not only to the spatial structures (intensity). In a subsequent short paper [2], the properties of the CS methods were analyzed in more depth. It was found that the spatial detail injected by CS methods features a low-pass component that has the effect of exactly overlapping the interpolated MS bands over the PAN image and a high-pass component that provides the actual sharpening. In MRA-based methods, the low-pass component that recovers the alignment and reduces aliasing is missing. If the CS method features an intensity component, like that defined in (1), the low-pass component is exactly given by the LS residue, ϵ, which can be calculated as the difference between $\mathbf{P}_L$ and $\widehat{\mathbf{I}}_L$, as shown in (1). The problem was to find an injection model suitable for the shift of both structures and colors. The multiplicative injection model, derived from the model of transfer of solar radiation through the atmosphere, was first proposed in [3].

In order to devise a correction of interpolated MS, we used the space-varying multiplicative gain [18], instead of the constant projection-based injection gain used in [2], to drive the injection of the space-varying residue, $\epsilon = \mathbf{P}_L - \widehat{\mathbf{I}}_L$, into the interpolated MS bands

$$\begin{aligned} \widetilde{\mathbf{M}}_k^* &= \widetilde{\mathbf{M}}_k + \frac{\widetilde{\mathbf{M}}_k}{\widehat{\mathbf{I}}_L} \cdot \epsilon = \widetilde{\mathbf{M}}_k \cdot \left(1 + \frac{\epsilon}{\widehat{\mathbf{I}}_L}\right) \\ &= \widetilde{\mathbf{M}}_k \cdot \left(1 + \frac{\mathbf{P}_L - \widehat{\mathbf{I}}_L}{\widehat{\mathbf{I}}_L}\right) = \widetilde{\mathbf{M}}_k \cdot \frac{\mathbf{P}_L}{\widehat{\mathbf{I}}_L} \end{aligned} \tag{4}$$

in which $\widetilde{\mathbf{M}}_k^*$ is the corrected version of $\widetilde{\mathbf{M}}_k$ that overlaps to $\mathbf{P}_L$. Equation (4) shows that the injection of the residue, ϵ, multiplied by the multiplicative model, that is, the ratio $\widetilde{\mathbf{M}}_k/\widehat{\mathbf{I}}_L$, into the interpolated MS band is equivalent to its multiplication by the ratio $\mathbf{P}_L/\widehat{\mathbf{I}}_L$.

The multiplicative injection gain, namely, $\widetilde{\mathbf{M}}_k/\widehat{\mathbf{I}}_L$, benefits from the correction of the atmospheric scattering term, called haze. Haze correction has been found to be effective for pansharpening [18], [19], [20], especially in vegetated areas. Specifically, the correction of atmospheric path radiance

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should be applied to the multiplicative injection gain by which the residue is modulated before being injected to correct the shift of the current MS band. Let h_k denote the haze term of the kth band. Let also $\widehat{w}_0$ approximate the haze term of $\widehat{\mathbf{I}}_L$. If we replace $\widetilde{\mathbf{M}}_k$ with $\widetilde{\mathbf{M}}_k - h_k$ and $\widehat{\mathbf{I}}_L$ with $\widehat{\mathbf{I}}_L - \widehat{w}_0$ in (4), we find that the haze-harmonized version of the interpolated kth band, $\widetilde{\mathbf{M}}_k$, will be

$$\widetilde{\mathbf{M}}_k^{h*} = \widetilde{\mathbf{M}}_k \cdot \frac{\mathbf{P}_L - \widehat{w}_0}{\widehat{\mathbf{I}}_L - \widehat{w}_0}. \tag{5}$$

Notice that (5) is expressed in the same units as (4), typically exoatmospheric spectral radiance (SR). If the data are in surface reflectance format, explicit dehazing is unnecessary. The same happens trivially if $\widehat{w}_0 \approx 0$.

B. Harmonization of WorldView-2/3 Data (4 + 4 + PAN)

Although never explicitly addressed, the set of MS bands in (1) is implicitly supposed to be aligned with each other. This condition was satisfied until the first satellite of the WorldView MS scanner generation was successfully launched in 2009 [21]. Whenever WorldView-2/3 data are concerned (WorldView-4, which is no longer operational since 2019, featured a unique MS instrument), the geometry of the view allows the eight spectral bands to be imaged from two different positions along the orbit, both different from PAN, as shown in Fig. 3(b). The lack of an accurate digital surface model (DSM) can cause local shifts between the old bands, blue (B), green (G), red (R), and near infrared 1 (NIR1), and the four bands imaged by the new MS instrument, coastal (C), that is, violet, yellow (Y), red edge (RE), and outermost NIR (NIR2).

The problem can be solved by considering the eight MS bands of WorldView-2/3 as produced by two separate instruments and by splitting the regression between the instruments, while in principle (1) might be written for all eight bands together. Let $S1$ and $S2$ denote the subscripts of the bands of the old and new MS scanners, respectively. Based on the sequence of the bands arranged by ascending wavelength, $S1 = \{2, 3, 5, 7\}$ and $S2 = \{1, 4, 6, 8\}$ [see Fig. 2(b)]. Equation (1) can be split into

$$\begin{aligned} \mathbf{P}_L &= \widehat{w}_{01} + \sum\nolimits_{k \in S1} \widehat{w}_k \cdot \widetilde{\mathbf{M}}_k + \boldsymbol{\epsilon}_1 \triangleq \widehat{\mathbf{I}}_{L1} + \boldsymbol{\epsilon}_1 \\ \mathbf{P}_L &= \widehat{w}_{02} + \sum\nolimits_{k \in S2} \widehat{w}_k \cdot \widetilde{\mathbf{M}}_k + \boldsymbol{\epsilon}_2 \triangleq \widehat{\mathbf{I}}_{L2} + \boldsymbol{\epsilon}_2 \end{aligned} \tag{6}$$

in which $\widehat{w}_k$, including the offsets $\widehat{w}_{01}$ and $\widehat{w}_{02}$, is the LS space-constant spectral weight and $\boldsymbol{\epsilon}_1$ and $\boldsymbol{\epsilon}_2$ are the space-varying residues of the matching of old and new instruments with the unique PAN.

In order to devise a correction of interpolated MS, we still used the space-varying multiplicative detail-injection gain, instead of the constant projection-based gain, to rule the injection of the space-varying residue, either $\boldsymbol{\epsilon}_1$ or $\boldsymbol{\epsilon}_2$, into the interpolated MS bands of the old and new MS scanners, respectively

$$\widetilde{\mathbf{M}}_{k,k \in Si}^{*} = \widetilde{\mathbf{M}}_{k,k \in Si} \cdot \frac{\mathbf{P}_L}{\widehat{\mathbf{I}}_{Li}} \tag{7}$$

in which $i \in \{1, 2\}$ and $\widetilde{\mathbf{M}}_k^*$ is the corrected version of $\widetilde{\mathbf{M}}_k$ that overlaps with $\mathbf{P}_L$. Analogously, with haze correction in the injection gain of the residue, we obtain the following:

$$\widetilde{\mathbf{M}}_{k,k \in Si}^{h*} = \widetilde{\mathbf{M}}_{k,k \in Si} \cdot \frac{\mathbf{P}_L - \widehat{w}_{0i}}{\widehat{\mathbf{I}}_{Li} - \widehat{w}_{0i}}. \tag{8}$$

The haze correction is expected to be more or less effective depending on the extent of the coefficients $\widehat{w}_{0i}$. Whenever they are close to zero, this is unnecessary.

IV. Assessment of Fusion Products

Since the seminal paper by Wald et al. [22] was published almost three decades ago, quality evaluation of pansharpened images has been the subject of extensive studies. Quality was recognized to change with the absolute scale at which fusion is performed and not only with the relative scale ratio of MS to PAN [23]. In fact, quality is crucial for instruments that have the blue channel and the PAN's bandwidth comprising the RE and part of the NIR wavelengths [1] because the NIR channel is not correlated with visible channels in the presence of vegetation [24]. This occurs because Landsat 7 ETM + has been equipped with a PAN band that encompasses part of the NIR wavelengths.

Quality evaluation plays a major role in the definition and development of MS pansharpening methods. The original protocol [22] foresees two separate checks.

1) *Consistency*: Checked at the spatial scale of the fusion product.
2) *Synthesis*: Checked at a spatial scale r times greater than that of the original PAN (r is the MS-to-PAN scale ratio).

In the past, only the synthesis property has usually been checked [25]. Later, it was recognized that consistency also plays a key role [26]. The main advantage of synthesis, which involves a reduced resolution (RR) test, is the accuracy in the performance assessment due to the presence of a reference image (ground truth, GT, equal to the original MS) and the availability of reliable and nonparametric similarity/dissimilarity measures [27], whose result cannot be changed by adjusting running parameters. The main drawbacks are related to the implicit assumption of scale invariance of the scene. Also, the bias introduced into the simulation procedure by the spatial filters, used for the spatial degradation of both the datasets and PAN filtering to obtain $\mathbf{P}_L$, will likely favor MRA rather than CS [28], because the unknown MTF, that is, the spatial frequency response of the imaging system, becomes identical to the frequency response of the separable 2-D filter used for spatial degradation.

These disadvantages can be overcome by resorting to a full resolution (FR) assessment. We recall that the synthesis property is not applicable to multimodal fusion [29], while consistency always holds. However, the consistency of Wald's protocol is a necessary condition, but not sufficient, unless it is coupled with the synthesis property [22], [26]. Wald's consistency is spectral, and an analogous spatial consistency measure was introduced one year later by Zhou et al. [30] to achieve a condition that is both necessary and sufficient. Unfortunately, Zhou's protocol presumably ignored Wald's protocol, and the

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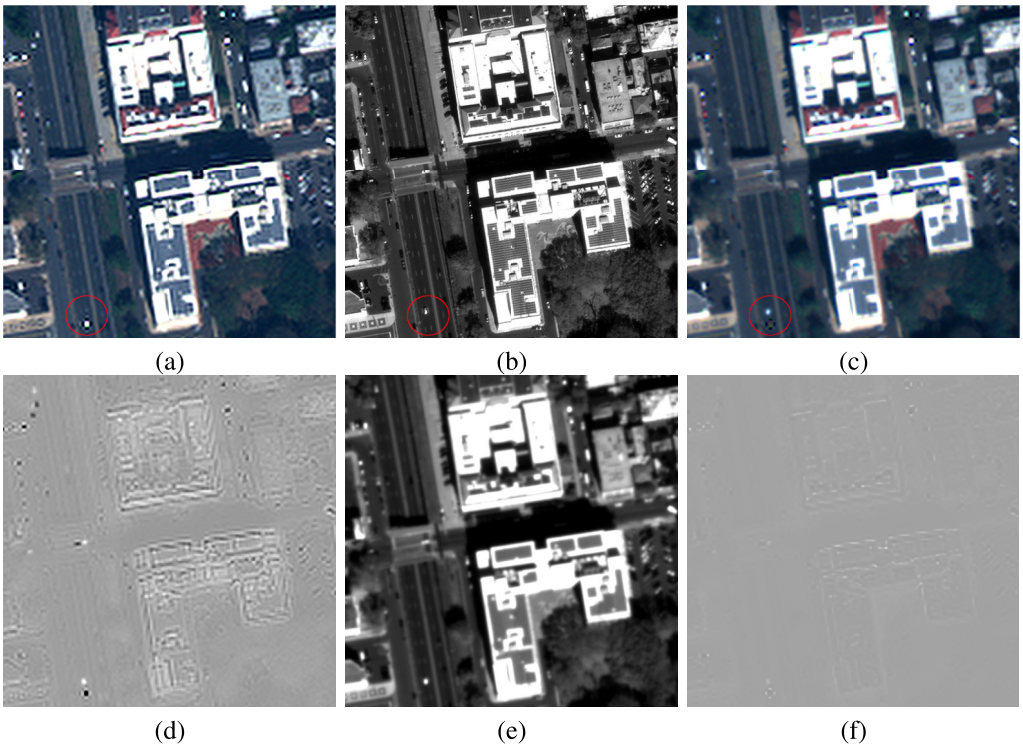


Fig. 4. 512 × 512 icon of test GeoEye-1 image. (a) Interpolated MS B3-B2-B1 (true color). (b) PAN. (c) Harmonized MS B3-B2-B1. (d) Residue of multivariate regression to low-pass-filtered PAN of resampled MS bands, $R^2 = 0.93914$. (e) Low-pass-filtered PAN. (f) Residue of harmonized MS bands to low-pass-filtered PAN, $R^2 = 0.99810$. Harmonization stands out in the moving vehicle surrounded by a red circle in the lower left part of (a)–(c).

spectral consistency that was proposed was flawed. However, in conceiving a counterpart in spatial consistency, it laid the foundations of the full-scale evaluation [31], [32].

The inconvenience of FR assessment is a reduction in the significance of evaluations because quality is not directly measured by similarity to GT, but inferred from indirect consistency measurements [26], [30], [31], [32], [33], [34] or extrapolated from RR measurements [23], [35]. Up-to-date studies [36] have suggested that the original MS image may not be suitable for fusion and especially for consistency measurements, due to the presence of aliasing patterns in MS, but not in PAN, and uncorrected local shifts, due to uncompensated parallaxes originating from different viewpoints along the orbit. Inconsistencies in FR protocols had previously been noted [34], [37], but the reasons were unclear. Here, the problem of quality measured on a full scale is tackled by resorting to harmonized MS and PAN datasets, for both fusion and assessment, through a proper benchmarking [38]. We wish to remark that in the present context, the term *quality* represents the *fidelity* to a hypothetically available reference and has no relationship with the intrinsic quality of the data produced by the instrument [39].

V. Experimental Results

A. GeoEye-1 Dataset

A GeoEye-1 test image was taken in the area of Trenton, NJ, USA, on September 27, 2019. The on-orbit limit sampling interval at the nadir (half the system resolution) is 1.64 m for MS and 0.41 m for PAN. The spatial sampling interval (SSI) of the resampled geocoded product is 2 m for MS and 0.5 m for PAN, resulting in a scale ratio, r, equal to four. The MS image features four spectral bands: blue (B), green (G), red (R), and near infrared (NIR), as shown in Fig. 2(a). The image size is 512 × 512 pixels for MS and 2048 × 2048 for PAN. The radiometric resolution of the packed DN format is 11 bits; the conversion coefficients to the top-of-atmosphere (TOA) SR format have been extracted from the metadata; the offsets are all equal to zero [40]. The true color compositions of the MS bands (B3, B2, and B1) and the PAN image are shown in Fig. 4(a) and (b), respectively.

Fig. 4(c) presents the harmonized MS image calculated as in (4), without haze correction. Compared to the resampled MS image in Fig. 4(a), the harmonized version is much more regular and similar to the low-pass-filtered PAN, shown in Fig. 4(e), from which it has been calculated. Fig. 4(d) reports the residue, ϵ, of the multivariate regression of the four interpolated MS bands toward the low-pass-filtered PAN, before the four bands have been harmonized. Apart from the blobs relative to moving vehicles on the left side and some aliasing effects in the parking lot on the right, the residue is mostly concentrated on the tops of buildings, whose height is presumably unspecified in the available DEM used for the geometric corrections. The map of residues of the harmonized version is shown in Fig. 4(f): apart from few details on the top of buildings, the residues, especially at street level, have almost disappeared. After applying the proposed harmonization procedure, the coefficient of determination of the multivariate regression increases from $R^2 = 0.93914$ to $R^2 = 0.99810$.

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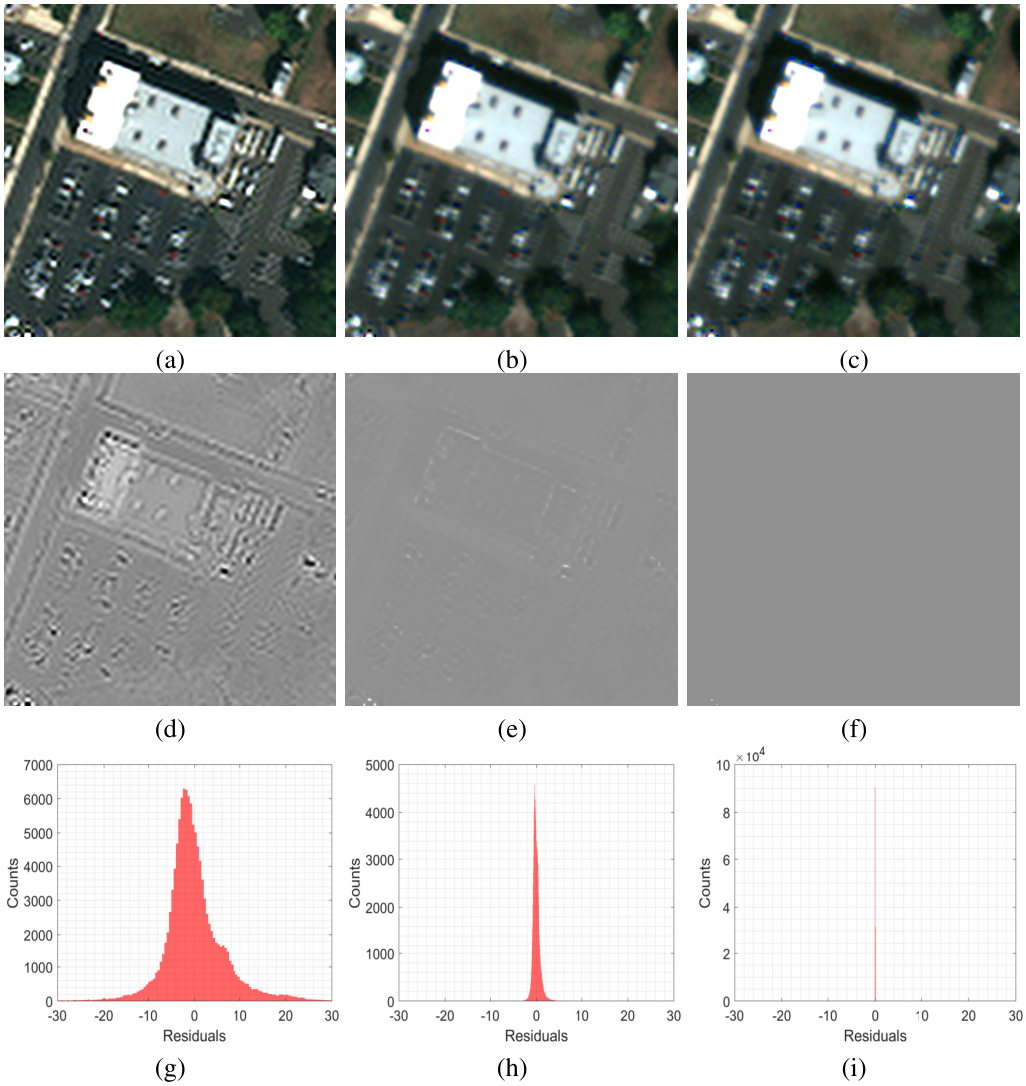


Fig. 5. 350 × 350 icon of test GeoEye-1 image. (a) Interpolated MS B3-B2-B1 (true color). (b) Harmonized MS B3-B2-B1. (c) Harmonized with haze MS B3-B2-B1. (d) Residue of multivariate regression to low-pass-filtered PAN of resampled MS bands, $R^2 = 0.94516$. (e) Residue of harmonized MS bands to low-pass-filtered PAN, $R^2 = 0.99892$. (f) Residue of haze-harmonized MS bands to low-pass-filtered PAN, $R^2 = 0.99994$. (g) Histogram of residue of multivariate regression to low-pass-filtered PAN of resampled MS bands, $\sigma = 7.12115$. (h) Histogram of residue of harmonized MS bands to low-pass-filtered PAN, $\sigma = 0.99791$. (i) Histogram of residue of haze-harmonized MS bands to low-pass-filtered PAN, $\sigma = 0.21735$.

To provide a deeper understanding of the characteristics of the harmonization process and quantify the benefits of dehazing, a 350 × 350 fragment, at the scale of PAN, of the same test image, showing a building having medium height and a parking lot with geometric lines to highlight aliasing artifacts, has been considered. Fig. 5 shows: first row, bicubically interpolated MS, harmonized MS (4) and harmonized MS with haze correction (5); second row, maps of the residue in the three cases; and third row, histograms of the residue in the three cases. The coefficient of determination starts from $R^2 = 0.94516$, moves to $R^2 = 0.99892$ after harmonization [without haze correction (4)], and reaches $R^2 = 0.99994$ with haze correction (5). The small dots of residue that survive after dehazed harmonization are presumably due to shadowing effects, on which the radiative transfer model, on which the dehazing relies, does not hold. The performance of harmonization is impressive. Apart from the few dots, the extremely low variance values of the residue witness the extremely low noisiness of PAN, achieved through TDI.

Once the MS bands have been harmonized, that is, overlapped onto the PAN image and cleaned from aliasing artifacts, simulations of MS pansharpening have been carried out. The goal is to check the benefits of harmonization for the main class of MRA, or spatial methods. Since [8], CS, or spectral methods are known to be insensitive to spatial impairments of the data, that is, aliasing and misregistration. The benchmarks consist of seven traditional methods and three popular methods that exploit variational optimization (VO), based on either modeling or learning. All methods are taken from the MS Pansharpening Toolbox, originally distributed from [28] onward. The three VO methods are implementations of the authors, each optimized for a specific satellite scanner. Including plain expansion (EXP) without detail injection, such methods are the following.

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Fig. 6. Visual results of fusion without harmonization. (a) PAN. (b) Expanded MS w/o harmonization. (c) BT w/o harmonization. (d) GS w/o harmonization. (e) BTH w/o harmonization. (f) GSA w/o harmonization. (g) BDSD w/o harmonization. (h) AWLP-H w/o harmonization. (i) MTF-GLP-FS w/o harmonization. (j) SR-D w/o harmonization. (k) TV w/o harmonization. (l) A-PNN-FT w/o harmonization. Note the position of the moving vehicle on the lower left part.

1) MS image interpolated with bicubic interpolation (EXP).
2) Brovey transfom (BT) [1].
3) Gram–Schmidt (GS) spectral sharpening method [1].
4) Optimized BT with haze (BT-H) correction [19], [41].
5) GS with adaptive (GSA) intensity [42].
6) Band-dependent spatial-detail (BDSD) injection [43].
7) AWLP with haze (AWLP-H) correction [44].
8) GLP with MTF filters and full-scale (MTF-GLP-FS) detail injection modeling [45].
9) Sparse representation of spatial details (SR-D) [46].
10) Fusion based on total variation (TV) optimization [47].
11) Advanced pansharpening with neural networks and fine-tuning (A-PNN-FT) [48].

The results of the fusion of the original (uncorrected) dataset are depicted in Fig. 6. What immediately stands out is that, notwithstanding the presence of MS-to-PAN shifts on the top of buildings, all CS methods (BT, BT-H, GS, GSA, and BDSD) yield clearer and sharper images than those of the two MRA methods, AWLP-H and MTF-GLP-FS, which are known to be unable to compensate for aliasing and local misalignment. The three VO methods, SR-D, TV, and A-PNN-FT, behave similar to methods based on MRA because they do not exploit the intensity component. Instead, once the images have been corrected following (5), the fused images are almost identical. Fig. 7 highlights that the two MRA methods and the three MRA-like methods recover their top performance, which had been reduced by the mismatch between the datasets. The haze correction is crucial in the presence of shadows; elongated blobs of blueish appearance are generated aside shadowed regions. Such impairments are strongly mitigated if the haze correction is skipped.

Tables I and II report the values of established full-scale quality indexes [34] measured on the fused datasets. Full-scale

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Fig. 7. Visual results of fusion with harmonization. (a) PAN. (b) Expanded MS with harmonization. (c) BT with harmonization. (d) GS with harmonization. (e) BTH with harmonization. (f) GSA with harmonization. (g) BDSD with harmonization. (h) AWLP-H with harmonization. (i) MTF-GLP-FS with harmonization. (j) SR-D with harmonization. (k) TV with harmonization. (l) A-PNN-FT with harmonization. Note the position of the moving vehicle on the lower left part.

TABLE I

GE-1 TRENTON: FULL-SCALE QUALITY INDEXES W/O HARMONIZATION FOR FUSION/ASSESSMENT

	QNR	D_λ	D_s	**HQNR**	D_λ^F	D_s	**FQNR**	D_λ^F	D_s^F	**RQNR**	D_λ^F	D_s^R
EXP	0.9222	0.0000	0.0778	0.8478	0.0806	0.0778	0.7131	0.0806	0.2243	0.8068	0.0806	0.1225
BT	0.8773	0.0257	0.0996	0.7656	0.1498	0.0996	0.8397	0.1498	0.0124	0.8495	0.1498	0.0009
GS	0.8937	0.0133	0.0942	0.7762	0.1431	0.0942	0.8285	0.1431	0.0331	0.8565	0.1431	0.0005
BTH	0.8792	0.0271	0.0964	0.7749	0.1424	0.0964	0.8467	0.1424	0.0127	0.8575	0.1424	3.29×10^{-5}
GSA	0.8634	0.0376	0.1029	0.7470	0.1674	0.1029	0.8203	0.1674	0.0149	0.8303	0.1674	0.0028
BDSD	0.9256	0.0195	0.0560	0.7799	0.1738	0.0560	0.7905	0.1738	0.0433	0.8188	0.1738	0.0089
AWLP-H	0.9125	0.0367	0.0528	0.9037	0.0459	0.0528	0.9175	0.0459	0.0384	0.9057	0.0459	0.0507
MTF-GLP-FS	0.8894	0.0609	0.0530	0.9038	0.0456	0.0530	0.9159	0.0456	0.0403	0.8973	0.0456	0.0598
SR-D	0.8770	0.0599	0.0672	0.9154	0.0187	0.0672	0.8805	0.0187	0.1027	0.7808	0.0187	0.2043
TV	0.9569	0.0151	0.0285	0.9020	0.0716	0.0285	0.8087	0.0716	0.1290	0.8651	0.0716	0.0682
A-PNN-FT	0.9375	0.0516	0.0115	0.9292	0.0600	0.0115	0.8839	0.0600	0.0597	0.8953	0.0600	0.0476

quality assessment means that there is no reference, as in RR assessment, and quality measurement is brought back to the fulfillment of the two consistencies between the fused image and the original MS and PAN [26]. It has recently been noticed

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TABLE II
GE-1 TRENTON: FULL-SCALE QUALITY INDEXES W/ HARMONIZATION FOR FUSION/ASSESSMENT

	QNR	D_λ	D_s	**HQNR**	D_λ^F	D_s	**FQNR**	D_λ^F	D_s^F	**RQNR**	D_λ^F	D_s^R
EXP	0.9018	0.0000	0.0982	0.8551	0.0518	0.0982	0.7534	0.0518	0.2054	0.8444	0.0518	0.1095
BT	0.8484	0.0829	0.0749	0.8825	0.0461	0.0749	0.9350	0.0461	0.0198	0.9518	0.0461	0.0022
GS	0.8854	0.0522	0.0658	0.8574	0.0822	0.0658	0.8926	0.0822	0.0275	0.9176	0.0822	0.0002
BTH	0.8523	0.0805	0.0731	0.8845	0.0458	0.0731	0.9359	0.0458	0.0192	0.9530	0.0458	0.0012
GSA	0.8384	0.0898	0.0790	0.8913	0.0323	0.0790	0.9469	0.0323	0.0215	0.9635	0.0323	0.0043
BDSD	0.9126	0.0454	0.0440	0.9136	0.0443	0.0440	0.9194	0.0443	0.0380	0.9522	0.0443	0.0037
AWLP-H	0.8729	0.0673	0.0641	0.9076	0.0303	0.0641	0.9514	0.0303	0.0189	0.9631	0.0303	0.0069
MTF-GLP-FS	0.8518	0.0896	0.0643	0.9066	0.0311	0.0643	0.9468	0.0311	0.0228	0.9583	0.0311	0.0109
SR-D	0.9511	0.0331	0.0163	0.9611	0.0230	0.0163	0.9400	0.0230	0.0379	0.9570	0.0230	0.0205
TV	0.9345	0.0306	0.0360	0.9239	0.0416	0.0360	0.8547	0.0416	0.1081	0.9110	0.0416	0.0494
A-PNN-FT	0.9575	0.0304	0.0126	0.9587	0.0291	0.0126	0.9144	0.0291	0.0582	0.9372	0.0291	0.0347



(a) (b) (c)

Fig. 8. 512 × 512 portion of test WorldView-2 image at 0.5 m. (a) PAN. (b) Interpolated B5-B3-B2 (true color). (c) Interpolated B6-B4-B1 (false color). Each image displayed is taken by a unique instrument without combining the spectral bands of different scanners.

that full-scale quality indexes are sensitive to misalignment of the original datasets [37]. The measure of spatial consistency is crucial in the presence of misalignment [33], depending on the class of methods, CS or MRA. The paradox is that the better appearance of CS is penalized in the consistency checks. When the MS bands are harmonized and hence overlapped to PAN, they become the new starting point for fusion, but also for carrying out consistency checks. Therefore, for uncorrected data, the quality reference is provided by the original MS bands that are fused; for corrected data, the new MS bands overlapped to PAN are implicitly the reference for quality assessment.

The entry labeled as EXP denotes the absence of injected details. In the case of no harmonization, the original data are simply resampled with bicubic interpolation. In the case of realignment, the corrected data are generated at the spatial scale of PAN, and resampling is implicit in the correction procedure. The values of the quality indexes for the fusion of uncorrected data in Table I reveal that all CS methods are poorer than the MRA methods, despite the fact that the visual appearance is exactly the opposite. The fusion of corrected MS data, with the latter used as a reference of consistency, turns the situation upside down; all the more advanced methods, except EXP, are pretty comparable to each other according to all indexes. The explanation of this is very simple. If CS methods force the fused image to overlap over PAN, they shift the fused image from the original MS, which is the reference for spectral consistency. Hence, the latter becomes poorer. Instead, with automatic harmonization, both the CS and MRA methods do not change the alignment of the data, which is the same as the alignment of the data that have been fused. Therefore, consistency measurements are not altered because all MS data, initial, fused, and reference, are aligned. Eventually, we cannot help noting that the index QNR, dating back to 2008, is moderately sensitive to shifts but has the drawback that the perfect spectral consistency of EXP conceals its poor spatial consistency; hence, EXP is always the best fusion method, which is puzzling. We wish to recall that each index is normalized in [0,1], but in such an interval, the mapping of scales is nonlinear. The indexes are suitable for relatively ranking methods, rather than for providing an absolute quality measure. We cannot help remarking the excellent performance of VO-based methods, even though their running parameters and offline training were calculated on data without harmonization.

B. WorldView-2 Dataset

A WorldView-2 image was acquired in the area of Sydney, Australia, on August 21, 2012. The SSI is 2 m for MS and 0.5 m for PAN. The study area is 2048 × 2048 PAN and 512 × 512 MS. Both MS and PAN have been orthonormalized using a DEM available at 10 m. Fig. 8 shows the 0.5-m PAN, the true color composition of the bicubically interpolated bands acquired by the MS1 instrument, and the false color composition of the interpolated MS2 bands. The

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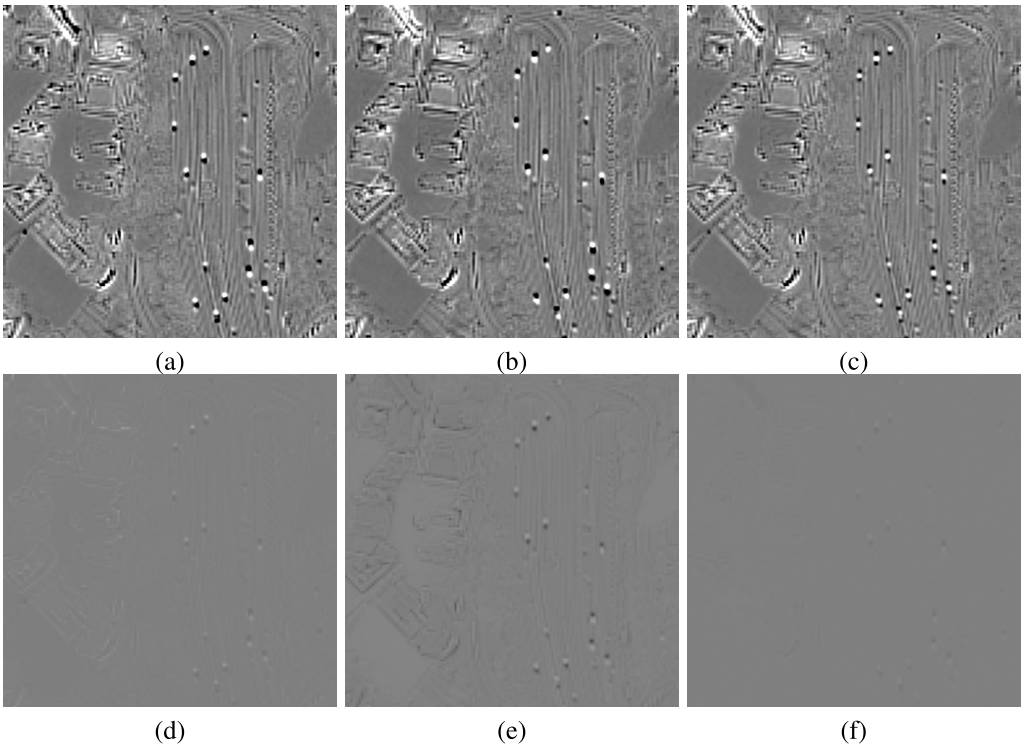


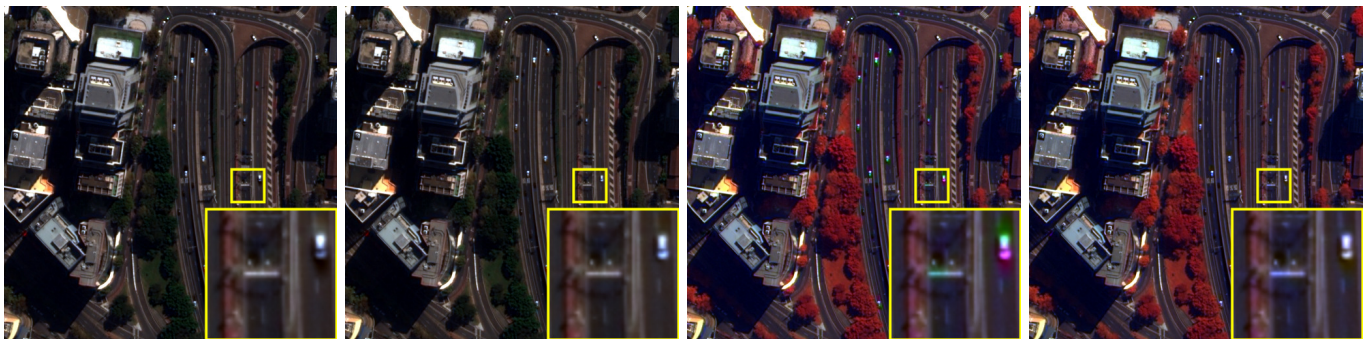
Fig. 9. Residue of multivariate regression to low-pass-filtered PAN of (a) four bands of MS1, R^2 = 0.97954, (b) four bands of MS2, R^2 = 0.97846, (c) eight bands together (MS1 + MS2), R^2 = 0.984490, (d) corrected bands of MS1, R^2 = 0.99993, (e) corrected bands of MS2, R^2 = 0.99968, and (f) MS1 + MS2 after correction, R^2 = 0.99999. All residue maps have been linearly stretched by the same factor for displaying convenience.

three icons display the data captured by each of the three instruments in Fig. 3. There is no misalignment of bands within the same bitmap, corresponding to a unique instrument, but there is misregistration between bands taken by different instruments. The roofs of the buildings are framed with slightly different perspectives, partially corrected during cartographic resampling. The displacement of vehicles traveling on the road indicates the order of acquisition.

Fig. 9 shows the map of the LS residue of the multivariate regression of MS to low-pass-filtered PAN and the related matching success, given by R^2, defined in (3). The residue is mostly concentrated in man-made structures, for which the DEM used for the orthonormalization was not of adequate spatial resolution. What immediately appears is the presence of moving vehicles along the main road, whose motion can be detected by looking at Fig. 8. The variance of the zero-mean residue is large for the match of MS2, slightly lower for MS1, which better encompasses PAN, while the outermost bands of MS2 lie outside the bandwidth of PAN. If regression is carried out on the corrected bands of (4), the residue almost vanishes: more for MS1 only, less for MS2 only, due to the different layouts of the bands of the two instruments. Using the eight corrected bands in (8), the regression achieves a value of R^2 of 0.99999, which is within the limits of the instrumental noise [49]. The few nonnull residues mostly appear on the road and originate from interpolation errors of moving cars. In this case, the haze correction is not useful because both the offsets, $\widehat{w}_{01}$ and $\widehat{w}_{02}$, are close to zero.

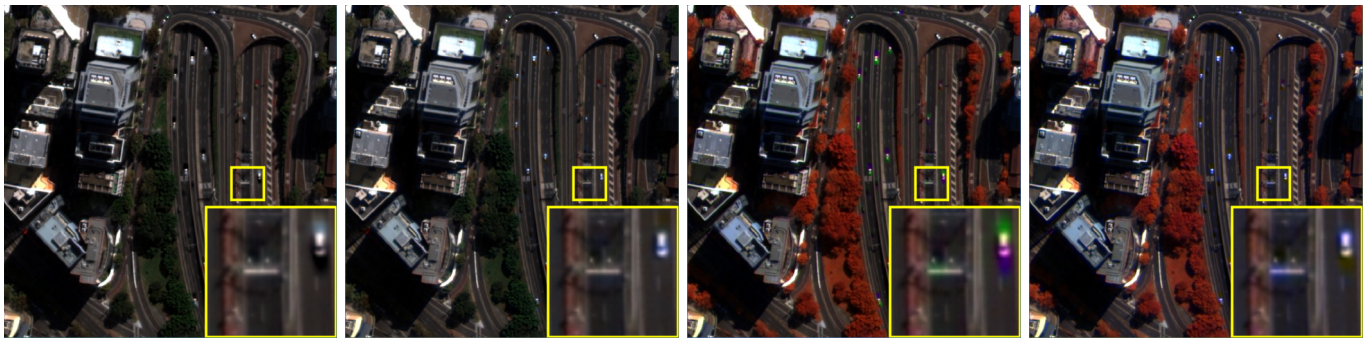
The benefits of harmonization of two separate MS scanners will be highlighted, focusing on moving vehicles. The comparison of four among the state-of-the-art methods used in Section V-A is reported in Figs. 10–13. Without correction, MRA methods are poorer than CS methods on moving cars, but out of the two CS methods, BT-H is better than GSA, thanks to its multiplicative injection model, compared to the projection model of GSA. With the full correction [see (8)], all fusion methods perform substantially well. Figs. 10 and 11 show two color compositions of the uncorrected and corrected MS data that have been fused, the eight bands together. When the fused data are displayed in true color, the data are taken by MS1 only and the shift toward PAN is almost completely absorbed by the CS methods (BT-H and GSA), analogously to what happened in [3]. The effect of the correction is hardly perceivable in the magnified detail of the car. When the bands displayed are taken from both instruments, the intersensor shift is not completely canceled by the unique residue calculated on all eight bands. The surviving misalignment is visible in the magnified box as a desynchronization of the color hues of the car. After applying the full correction, the alignment is restored. The problem originating from the presence of two separate MS scanners, imaging from different parallaxes, is that the alignment of MS over PAN cannot be implicitly recovered by CS pansharpening methods, as happens with only two onboard instruments, as in the case of GeoEye-1. The situation is different for the two MRA methods reported in Figs. 12 and 13. Here, the uncorrected true color is poor and the uncorrected false color is extremely poor. Thanks to the correction, in the challenging case of inter-instrument color composition, the bleeding of colors is restrained, with acceptable results for all methods. We wish to recall that

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(a) (b) (c) (d)

Fig. 10. BT-H fusion results. (a) B5-B3-B2 composition w/o harmonization, (b) B5-B3-B2 w/ harmonization, (c) B6-B3-B1 w/o harmonization, and (d) B6-B3-B1 w/ harmonization.



(a) (b) (c) (d)

Fig. 11. GSA fusion results. (a) B5-B3-B2 composition w/o harmonization, (b) B5-B3-B2 w/ harmonization, (c) B6-B3-B1 w/o harmonization, and (d) B6-B3-B1 w/ harmonization.



(a) (b) (c) (d)

Fig. 12. AWLP-H fusion results. (a) B5-B3-B2 composition w/o harmonization, (b) B5-B3-B2 w/ harmonization, (c) B6-B3-B1 w/o harmonization, and (d) B6-B3-B1 w/ harmonization.

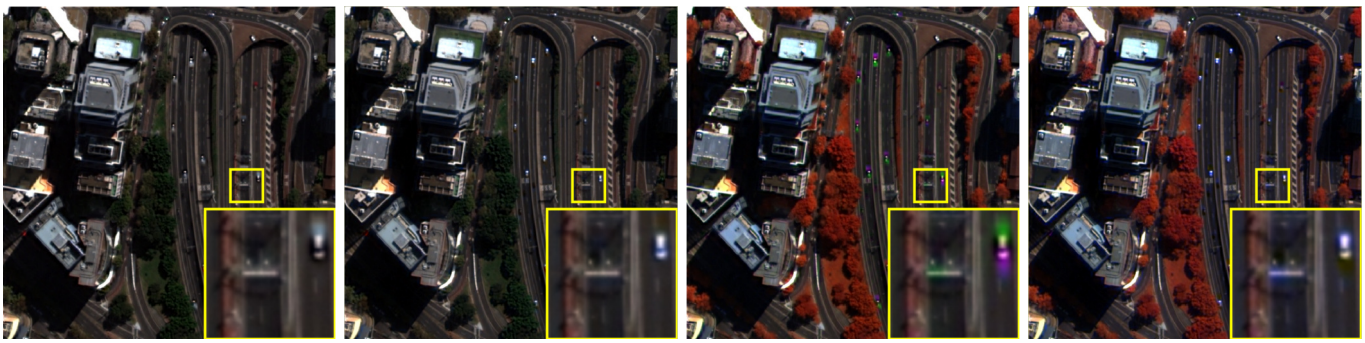
according to the proof stated in [8], only moderate shift can be recovered. The high speed of cars on the main road originates shifts of MS toward PAN up to 5 and 6 pixels at the scale of PAN. Such a shift cannot be well approximated by the first-order development of the mathematical function representing the object, the vehicle, in the present case.

The quality scores of all the fusion methods reported in Tables III and IV match those of the GeoEye-1 dataset. Here, CS methods are slightly less penalized without harmonization, and RQNR tends to saturate its values, also without harmonization. The presence of two separate MS scanners penalizes also CS methods, though to an extent lower than MRA ones.

VI. Discussion

After presenting the results on the two datasets of two different instruments, in which the acquisition geometry is progressively complicated by the presence of two and then three separate instruments, a series of considerations can be made. For scanners with decametric resolution, e.g., Sentinel-2 and the operational land imager (OLI), the problem of misalignment does not exist, either with a unique instrument capturing all the MS bands (Sentinel-2) or also with three separate instruments (OLI) because the pixel size, SSI, is comparable with the spatial scale at which the DEM is generally

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(a) (b) (c) (d)

Fig. 13. MTF-GLP-FS fusion results. (a) B5-B3-B2 composition w/o harmonization, (b) B5-B3-B2 w/ harmonization, (c) B6-B3-B1 w/o harmonization, and (d) B6-B3-B1 w/ harmonization.

TABLE III

WV-2 SYDNEY: FULL-SCALE QUALITY INDEXES W/O HARMONIZATION FOR FUSION/ASSESSMENT

	QNR	D_λ	D_s	**HQNR**	D_λ^F	D_s	**FQNR**	D_λ^F	D_s^F	**RQNR**	D_λ^F	D_s^R
EXP	0.9287	0.0000	0.0713	0.8999	0.0311	0.0713	0.7228	0.0311	0.2540	0.8586	0.0311	0.1138
BT	0.8623	0.0544	0.0881	0.8528	0.0648	0.0881	0.8986	0.0648	0.0391	0.9341	0.0648	0.0012
GS	0.9255	0.0227	0.0530	0.8769	0.0740	0.0530	0.8819	0.0740	0.0476	0.9252	0.0740	0.0009
BTH	0.8607	0.0559	0.0884	0.8692	0.0465	0.0884	0.9178	0.0465	0.0374	0.9535	0.0465	1.53×10^{-6}
GSA	0.8511	0.0604	0.0942	0.8609	0.0495	0.0942	0.9127	0.0495	0.0397	0.9503	0.0495	0.0002
BDSD	0.9506	0.0131	0.0368	0.8977	0.0680	0.0368	0.9016	0.0680	0.0327	0.9320	0.0680	1.54×10^{-5}
AWLP-H	0.8902	0.0451	0.0678	0.9150	0.0185	0.0678	0.9505	0.0185	0.0316	0.9677	0.0185	0.0140
MTF-GLP-FS	0.8610	0.0646	0.0796	0.8932	0.0295	0.0796	0.9445	0.0295	0.0268	0.9512	0.0295	0.0199
SR-D	0.9278	0.0392	0.0344	0.9504	0.0157	0.0344	0.9372	0.0157	0.0479	0.9392	0.0157	0.0459
TV	0.9332	0.0263	0.0416	0.9315	0.0282	0.0416	0.9579	0.0282	0.0143	0.9344	0.0282	0.0385
A-PNN-FT	0.9506	0.0170	0.0329	0.9377	0.0303	0.0329	0.9328	0.0303	0.0380	0.9529	0.0303	0.0173

TABLE IV

WV-2 SYDNEY: FULL-SCALE QUALITY INDEXES W/ HARMONIZATION FOR FUSION/ASSESSMENT

	QNR	D_λ	D_s	**HQNR**	D_λ^F	D_s	**FQNR**	D_λ^F	D_s^F	**RQNR**	D_λ^F	D_s^R
EXP	0.9384	0.0000	0.0616	0.9101	0.0301	0.0616	0.7550	0.0301	0.2215	0.8790	0.0301	0.0937
BT	0.8639	0.0620	0.0790	0.8906	0.0331	0.0790	0.9265	0.0331	0.0418	0.9662	0.0331	0.0007
GS	0.9287	0.0270	0.0455	0.9016	0.0554	0.0455	0.8996	0.0554	0.0477	0.9436	0.0554	0.0010
BTH	0.8600	0.0656	0.0796	0.9074	0.0141	0.0796	0.9466	0.0141	0.0398	0.9859	0.0141	7.69×10^{-11}
GSA	0.8625	0.0611	0.0814	0.9053	0.0144	0.0814	0.9449	0.0144	0.0413	0.9856	0.0144	1.64×10^{-7}
BDSD	0.9371	0.0202	0.0436	0.9162	0.0420	0.0436	0.9143	0.0420	0.0456	0.9566	0.0420	0.0015
AWLP-H	0.8809	0.0528	0.0700	0.9174	0.0135	0.0700	0.9530	0.0135	0.0339	0.9861	0.0135	0.0004
MTF-GLP-FS	0.8629	0.0679	0.0743	0.9046	0.0227	0.0743	0.9369	0.0227	0.0413	0.9672	0.0227	0.0103
SR-D	0.9316	0.0259	0.0436	0.9459	0.0110	0.0436	0.9736	0.0110	0.0157	0.9821	0.0110	0.0071
TV	0.9311	0.0258	0.0443	0.9384	0.0182	0.0443	0.9549	0.0182	0.0274	0.9721	0.0182	0.0100
A-PNN-FT	0.9621	0.0104	0.0278	0.9521	0.0206	0.0278	0.9424	0.0206	0.0378	0.9654	0.0206	0.0143

available: 10 m. If the scale of the available DEM is 10 m and the scale of PAN is less than 1 m, it becomes impossible to correct the parallaxes with sufficient accuracy. In the presence of discontinuities of the ground surface, the registration error turns out to be a fraction of the scale of the available DEM, say 25%, with an optimistic view. Thus, a misalignment of up to three pixels of PAN is produced, spread over the height discontinuities of the surface. This problem has been recognized since full-scale quality assessments have become widespread. With fusion and assessment performed on spatially degraded data, typically by a factor of four, all pixel shifts are divided by four and do not produce appreciable misalignment effects. This is the reason why the proposed approach regards datasets that are at least VHR, corresponding to 1 m PAN.

Once the resampled MS bands have been automatically and accurately overlaid onto the PAN image, established MRA pansharpening algorithms recover their original top performance, which had been diminished in the passage from HR to VHR and EHR datasets. Performance is intended both in the visual sense and also according to widespread statistical quality indexes based on consistency measurements of fusion products toward unfused MS and PAN. In other words, the same resampled MS bands corrected for aliasing and shift compensation should be used as input to the fusion and quality assessment procedures.

According to the theoretical development reported in [8], the proposed preprocessing patches are intrinsically capable of removing possible aliasing artifacts, generated by an

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insufficient sampling step size, from the resampled MS bands, provided that the PAN image is aliasing-free. This is true because the PAN image is taken by sampling the same MTF in Fig. 1 with a step size four times greater. A question that may arise is the following: is the aliasing cancellation feasible also where MS and PAN are shifted? The answer is yes, because in [8], it is proven that suppression of the aliasing term is direct, namely, aliasing is canceled by injecting PAN minus the intensity component, which is a linear combination of the MS bands and overlaps to them. Conversely, in the framework of MRA fusion, aliasing can be indirectly canceled by means of PAN, using the analysis filter of PAN—as a limit case, an all-pass filter to achieve a nonredundant representation, suitable for data compression [50]—to create aliasing patterns that cancel those of the MS bands. This is an asset of Laplacian pyramids for fusion [51], [52]. However, for indirect cancellation, perfect overlap of MS and PAN is required [53].

It should be noted that the preprocessing patches are independent of the measure unit of the data, e.g., TOA SR, TOA reflectance, surface reflectance, in floating-point or packed fixed-point formats, due to multivariate regression, whose degree of matching, namely, R^2, does not change with the data units or formats [40], which implies that the datasets are harmonized, regardless of their initial formats, and the corrected datasets retain their original formats.

The proposed method does not require either training datasets or ancillary data (DEM, ground control points, and so on) and may work on a unique MS + PAN image. Last but not least, its computational complexity is extremely low, comparable to that of pansharpening methods that exploit multivariate regression, such as GSA [42] and BT-H [41]. For PAN and interpolated four-band MS of size 2048×2048: approximately 2.5 s; for PAN and interpolated eight-band MS of the same size: approximately 4.5 s. Regression employs the MATLAB function "fitlm," which is found in the "Statistics and Machine Learning Toolbox." Computing platform: 64-bit Windows 11 OS; Intel N150 processor at 800 MHz; 16-GB RAM.

Harmonization can be useful for a series of inconveniences: 1) mitigation of chromatic aberration, which produces shifts that increase linearly with the distance from the focal center among bands framed with different wavelengths; 2) elimination of systematic shifts generated by improper interpolation [16] of the MS dataset, including nearest-neighbor resampling; 3) fine alignment of spectral bands captured by optical payloads mounted on drones, in the presence of relief of terrain; and 4) estimation of the signal-dependent noise model of electrooptic instruments [54], [55]. The last item highlights that the residue encompasses all dissimilarities between the harmonizing and harmonized datasets, including the stochastic noise. In this sense, it is known that fusion methods based on nonseparable MRA [56] prevent the noise from being injected as if it were a structured detail.

VII. Conclusion and Developments

The study presented a simple idea that led to fully automated real-time data harmonization procedures. One is suitable for the MS and PAN datasets produced by systems featuring two separate onboard instruments, like GeoEye-1 and Pléiades, among systems that are currently operative. Another, for the MS1, MS2, and PAN datasets produced by the three instruments onboard WorldView-2/3. The space-varying residue of the multivariate linear regressions between the interpolated MS bands and the low-pass-filtered PAN is the key to detect and correct aliasing artifacts and local displacements. Simulations performed on a four-band GeoEye-1 image and an eight-band WorldView-2 image, both having 2-m MS and 50-cm PAN, show that local shifts up to at least 2 m are compensated, both in geometry and in colors.

The proposed strategy relies on the unimodality of MS and PAN instruments, by which the low-pass-filtered PAN image can be synthesized as a weighted combination of the four interpolated MS bands of each MS scanner, if there are more than one. The spatial residue of the multivariate regression locally measures the differences (aliasing and misalignment) between the intensity component of the MS bands captured by one instrument and the low-pass-filtered PAN. The correction consists of pixel-by-pixel multiplying the interpolated MS band by the ratio between the low-pass-filtered PAN and the synthetic LS intensity component. Correction of the atmospheric path radiance term (haze), whenever applicable, is beneficial in terms of reduction of the regression residue and of harmonization of the datasets but may introduce spectral artifacts on the edges of shadow regions. This issue deserves further investigation. However, haze correction can be enabled and disabled on user request.

Given the excellent performance of the unique learning-based method that has been considered, it would be interesting to investigate the behaviors of learning-based fusion when trained on harmonized MS and PAN data. In fact, there are methods that are capable of learning and compensating for small shifts [57], [58], as happens to all CS methods that operate without learning, but nobody has ever checked the improvements in learning carried out on perfectly aligned MS and PAN datasets.

The three-instrument harmonization procedure can be extended to a fourth onboard instrument: the SWIR MS scanner mounted on WorldView-3. The two-instrument procedure may be utilized for the fine co-registration of hyperspectral (HS) and PAN, where applicable, that is, HS and PAN mounted on the same platform; in addition to fusion, this issue is relevant for a spectral unmixing driven by PAN [59]. Other possible fields of application are landcover change detection [60] and landcover classification [61]. Once the MS data are resampled, harmonized, and then decimated, the aliasing artifacts on the native scale of the MS data will disappear, with expected benefits on temporal change and thematic classification maps.

The subsequent step is the multiplatform extension of the harmonization procedures. For single-platform harmonization, the radiometric consistency between the datasets is implicit. In the multiplatform case, instrumental and possibly atmospheric corrections may become crucial, while for the single-platform harmonization, the problem is only geometric. So far, geometric consistency has generally been assumed between the datasets of the two platforms, e.g., for applications of MS-HS fusion, where the radiometric consistency can be borrowed from the harmonized dataset [62]. The inconveniences that

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arise if the platforms acquire with different view angles, for example, "nadir" and "off-nadir," seem easily containable, thanks to the robust regression that forces the residual to correct the view angle of the "harmonized" dataset.

A necessary condition for the success of the harmonization procedures is the homogeneity of the data. However, if it is not possible to harmonize optical data with synthetic aperture radar (SAR) data, a sort of harmonization procedure can in principle be devised for interferometric SAR (InSAR) data [63], harmonizing single-look complex (SLC) pairs before calculating their normalized interferograms.

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